

wisdom in Latin America is that anti-inflation programs using the exchange rate as a nominal anchor (exchange-rate-based stabilization, or ERBS, programs) have been able to reduce the inflation rate without causing the initial output losses associated with programs that use some monetary aggregate as a nominal anchor (money-based stabilization programs).⁴

Of course, a theory for stabilization programs is not the same as a theory for the business cycle. But there should be some consistency among them. For example, a finding that nominal shocks do not have important real effects during Latin American stabilization programs would make it harder to maintain the monetarist view that such factors may have been important at any other point of the business cycle. And this is precisely what we find problematic: a reexamination of the evidence on ERBS programs shows that it is far from clear that the adoption of the exchange rate as a nominal anchor has been responsible, as the literature claims, for the economic fluctuations observed during those programs.

Figure 1 illustrates the consumption growth rates for the ten ERBS programs studied by Végh (1992). The vertical line indicates the year or quarter in which the ERBS program started.⁵ Casual inspection of the plots suggests that only in the first four cases did consumption experience the upward jump that theory predicts should occur upon announcement of ERBS programs.⁶ However, this theoretical prediction did not materialize in the remaining six cases. In particular, in none of these six did consumption grow faster than in the immediately preceding period. Instead, in four of the six cases, consumption growth was basically the same immediately before and immediately after the announcement of the corresponding ERBS program. In two of the four, the so-called consumption boom preceded the announcement. In the other two, there was no consumption boom whatsoever: consumption continued falling at approximately the same rate as before the ERBS programs began. Furthermore, in the last two cases, the ERBS program was followed instead by a consumption bust.

Therefore, the timing, intensity, or direction of consumption growth for the countries in Végh's study, after most ERBS programs began, appears to differ from that implied by the ERBS theory.

In this sense, at least four of the plots in Figure 1 (Chile, February 1978; Argentina, December 1978; Argentina, June 1985; and Israel, July 1985) could be interpreted using the non-

monetarist approach: the dynamics of output immediately after the announcement of an ERBS program were mere continuations of upswings or downturns that had begun earlier. In these four cases, forces other than the adoption of a fixed or pegged exchange rate were already driving the business cycle when the ERBS programs began. But such conclusions from the casual reading of two-dimensional plots would be premature.⁷ We are more persuaded, instead, by the more thorough empirical effort of Rebelo and Végh (1995), who conclude that monetarist-inspired theoretical models of ERBS programs are quantitatively incapable of replicating any significant fraction of the economic fluctuations associated with such programs.

The evidence on ERBS programs, both from casual plot readings and from the work of Rebelo and Végh, poses a serious challenge to monetarist theories of Latin American business cycles: if nominal exchange rate shocks in Latin America seem to have failed to produce the noticeable and consistent effects on consumption and other real variables predicted by monetarist-inspired theories precisely when they were given the best shot at it, how could they have significant real effects at other times?⁸

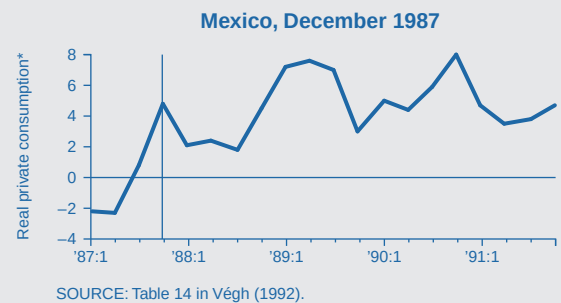
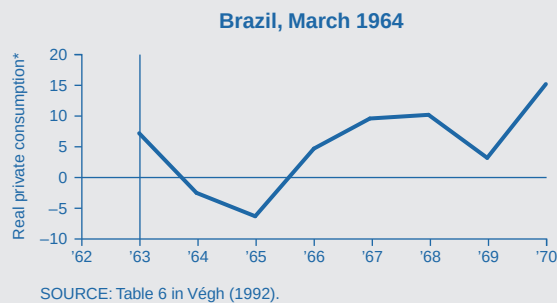
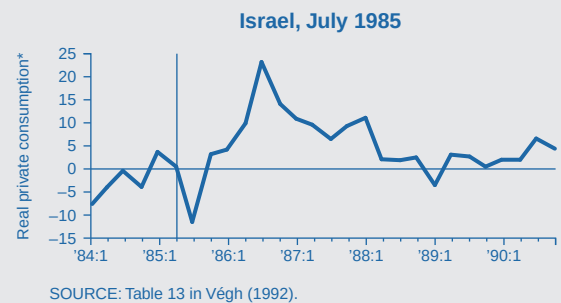
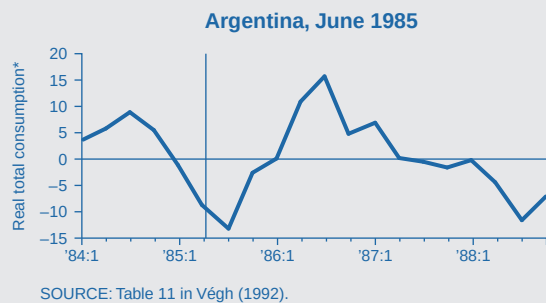
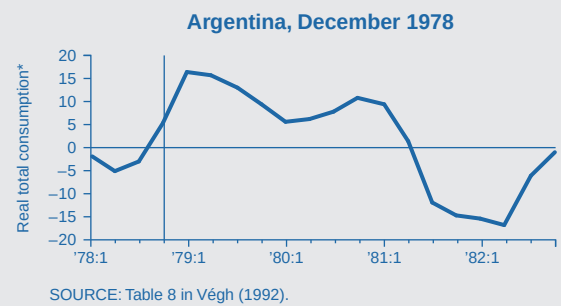
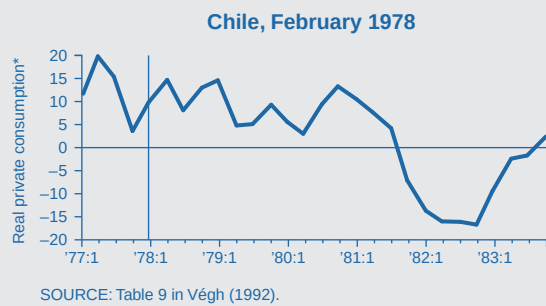
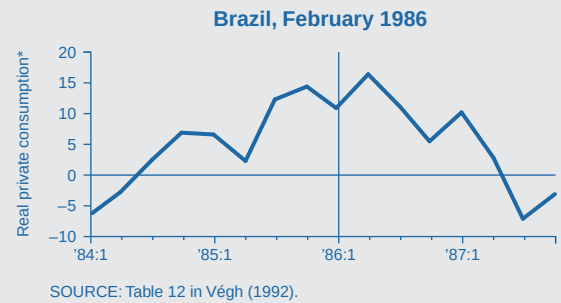
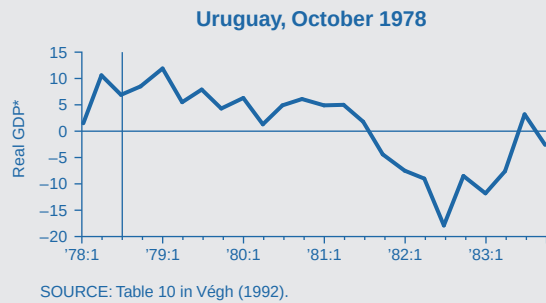
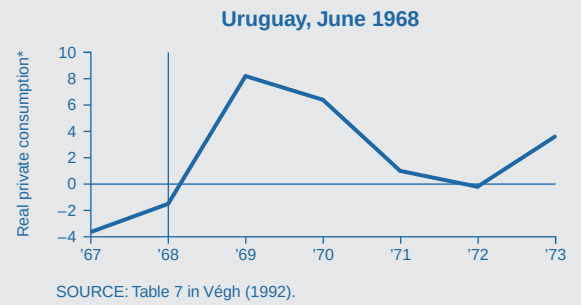
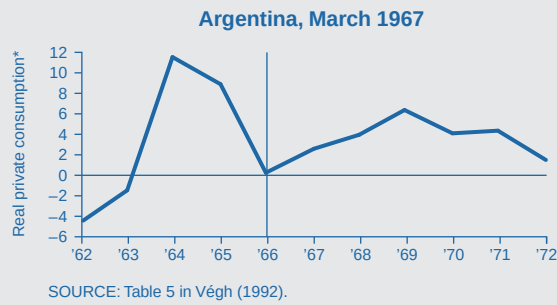
A natural next step in the research agenda is to pay more attention to real shocks as a potentially important source of the economic fluctuations observed in Latin American countries, including fluctuations observed during inflation stabilization programs.⁹ In principle, there is no reason the assessment of the quantitative importance of such shocks in Latin America could not be accomplished with the same kind of dynamic stochastic general equilibrium models the RBC tradition has used to that effect for the United States and other developed countries.

But such a research program must start by describing the data with a systematic, atheoretical methodology.¹⁰ The remaining sections of this article make a modest attempt in that direction by describing the business-cycle regularities of Argentina without imposing theoretical priors to the data.¹¹

Business-cycle regularities for Argentina

Some caveats about the data. National account data in Latin America are not as reliable as their U.S. and OECD counterparts.¹² In fact, because of frequent methodological changes and corrections of previous errors, the reported series may change substantially from one national account estimate to the next. This is indeed the case for Argentina. For example,

Figure 1
ERBS Programs



* Growth in percent with respect to the same period of the previous year.

volatility of consumption relative to output is substantially lower in the national accounts estimate at 1986 prices (released at the end of 1996) than in the previous estimates at 1970 prices.

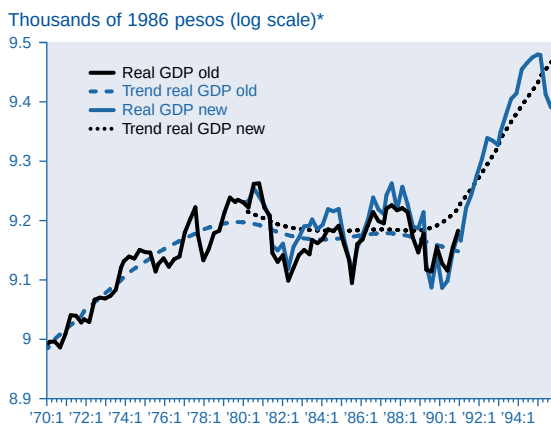
The example above emphasizes that in dealing with countries such as Argentina, researchers should heed the usual warning to appropriately weigh the quality of the data before taking for puzzles anomalies that in reality may be mere statistical artifacts. For that reason, we report the business-cycle regularities obtained from using two different estimates of GDP and its components. The comparison of the results from each data set will eventually give some idea of the confidence one should place on the business-cycle regularities of Argentina reported here or elsewhere (for examples, see Kaufman and Sturzenegger 1996 and Carrera, Féliz, and Panigo 1996).

One estimate (the “old” estimate), in constant prices of 1970, covers the 1970:1–90:4 period and was prepared by the Central Bank of Argentina. We obtained this estimate from the FIEL (Fundación de Investigaciones Económicas Latinoamericanas) data bank. The other estimate (the “new” estimate), in constant prices of 1986, covers the 1980:1–95:4 period. The figures for this estimate were taken from the publication *Oferta y Demanda Globales, 1980–1995*, prepared by the Dirección Nacional de Cuentas Nacionales. Notice that these two estimates overlap only during the 1980:1–90:4 period.¹³

Methodology. We characterize the business-cycle regularities of Argentina using Kydland and Prescott (1990) as a guide. Their procedure is inspired by Lucas (1977), who defines the business-cycle component of a variable as its deviation from trend. Kydland and Prescott define the trend of a variable as that which results from applying the Hodrick–Prescott filter (HP filter) to the raw data. Informally, this filter produces trends that are “close to the one that students of business cycles and growth would draw through a time plot” (Kydland and Prescott 1990).¹⁴ Application of the HP filter to Argentinean GDP, for example, produces the trend represented by the smoother curves in Figure 2.^{15,16}

Except for net exports, all variables in the tables of this article are expressed in natural logarithms, as is standard in the business-cycle literature.¹⁷ Since it is not possible to compute the logarithm of negative values, variables that can take on such negative values, such as net exports, were expressed instead as ratios to GDP. All the variables were seasonally adjusted using the X-11 procedure.

Figure 2
Real GDP, Old and New Estimates



* For visual effect, the old estimates have been rescaled so that their level is the same as for the new estimates in 1980:1.

SOURCES: Dirección Nacional de Cuentas Nacionales for new estimates; FIEL for old ones; authors' calculations for trends.

The tables report statistics that measure (1) the direction of the movements of a variable compared with that of real GDP (*procyclical*, in the same direction; *countercyclical*, in the opposite direction; *acyclical*, when there is no clear pattern); (2) the degree to which the variable follows the movements of real GDP (contemporaneous correlation); (3) the amplitude of fluctuations (volatility or relative volatility); and (4) the phase shift—that is, whether a variable changes before or after real GDP does (leads or lags the cycle, respectively.)

The statistics *volatility* corresponds to the standard deviation of the percentage by which the cyclical component of a variable deviates from trend. The statistics *relative volatility* is the ratio between the volatility of the variable of reference and the volatility of real GDP.

Real facts for Argentina

Output and its components: GDP. Table 1 reports statistics for real GDP and its major components. The first striking feature of the table is the high volatility of real GDP. According to the new national account estimates, the percentage standard deviation from trend of Argentina's real GDP is roughly 2.5 times larger than for the United States. Real GDP volatility is also high in the old national account estimates, but within the range observed in European countries such as Greece (2.85), Portugal (3.05), and Luxembourg (3.2).^{18,19}

Total consumption. An important caveat in interpreting the consumption evidence is that in Argentina's national account, consumption is computed as a residual, which casts consider-